



Slow wave dynamics of scalp EEG can be explained by simple statistical models of long-range connections

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ABSTRACT

Scalp-recorded electroencephalography (EEG) is thought to be driven by both local and global oscillations dependent on the cognitive state and task of the individual. However, many EEG studies assume that the activity is local, especially when inverse modeling EEG activity. In this work, we show that a simple model of purely macroscopic connections derived from biologically plausible distributions of long-range axon delays can drive many of the traditional features of scalp-recorded EEG dynamics. All that is required is a simple linear model of time delays in a linear vector autoregressive framework with a few parameters. We make several simplifying modeling assumptions in the model: only long-range excitatory connections are included, local activity is treated as stochastic noise, and nonlinear synaptic dynamics are omitted. As a proof of concept, we restrict the model to five broad brain regions (frontal, parietal, occipital, temporal, thalamic) and model resting state EEG with no external input. We show how this simple connection model is derived from theoretical principles of synaptic activity. The model is able to replicate many features of real EEG data, including resting-state alpha power and coherence (8–13 Hz). We show that model parameters can also be informed by empirical work on structural connectivity, axon diameter estimation, and functional connectivity of fMRI BOLD measures. However, some features of the macroscopic simulations are not ideal as a model for all features of resting EEG, such as high coherence in low-frequencies in the simulation as opposed to real data. Overall, the results support the explanation of many classical EEG findings in terms of macroscopic network behavior as opposed to local activity.

1. Introduction

Scalp-recorded EEG is known to be generated by the synchronous synaptic activity of neural action (Buzsáki, 2006; Nunez and Srinivasan, 2006). However, there is a diversity in the possible underlying computational mechanics that give rise to the synchronous synaptic activity (Breakspear, 2017; Glomb et al., 2022). Different theoretical ideas about EEG generation, instantiated by computational models in simulation, abound. These computational models can be split into two general categories: *local* and *macroscopic* (Nunez and Srinivasan, 2006; Glomb et al., 2022). Local models could further be split into *microscopic* and *mesoscopic* models (Glomb et al., 2022). However, an open question remains about which spatial scale is the most useful description for certain EEG features. In this work, we derive, simulate from, and compare EEG data to a simple macroscopic model that relies only on structural and functional connectivity. The simple linear statistical model can describe features of EEG recordings from participants at

rest and likely could be extended to describe other features of EEG recordings.

Our macroscopic model of EEG assumes that long-range white-matter axons are excitatory and carry electrical signals between regions of the cortex and the thalamus (Nunez and Srinivasan, 2006). Possible additional contributions of local delay mechanisms (e.g., rise and fall times of postsynaptic potentials) are neglected here to focus solely on the influence of macroscopic delays. In order to model the broad dynamics of the EEG, we only assume five macroscopic regions in our model: frontal, parietal, occipital, temporal, and thalamic. We show that the classical properties of scalp EEG can be explained by the long-range excitatory dynamics between these regions.

1.1. Differentiating macroscopic dynamics from mesoscopic dynamics

Many researchers have previously simulated human electrophysiology from populations of single neuron models. For instance, Izhikevich

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(2003) combined the famous Hodgkin–Huxley model with integrate-and-fire neurons to simulate interconnected networks of neurons. These models can be further extended to the mesoscopic scale in order to model oscillatory dynamics. For instance, there is work on the mesoscopic scale to model beta (13–30 Hz) and gamma (30–80 Hz) frequencies based on inhibition-based dynamics (Whittington et al., 2000).

There is also a rich history of modeling human electrophysiology, including explaining observed EEG based on population dynamics with the Wilson–Cowan model and extensions (Wilson and Cowan, 1973; Srinivasan et al., 2013). Wilson–Cowan models exhibit intrinsic oscillatory activity due to the interactions of excitatory and inhibitory neural populations. These types of models are very powerful for predicting oscillations, typically >20 Hz, at certain spatial locations in the cortex that give rise to fast wave dynamics of EEG on the scalp. Many mesoscopic models have components that act as “pacemakers” to produce oscillations, which could be due to local or non-local neurons or populations of neurons (e.g., Whittington et al., 2000). Extensions of these models that include delays for long-range brain communication can give rise to synchrony in these oscillations (Sanz-Leon et al., 2015; Finger et al., 2016; Breakspear, 2017; Glomb et al., 2022).

However, our model does not explicitly contain neurons or populations of neurons to generate oscillations. Rather, our model relies only on macroscopic connectivity with delays and the synchrony of input arriving at cortical locations from long-range axons. Similar global models based on axon propagation delays have been around at least since the 1970s (Nunez, 1974, 1989; Nunez and Cutillo, 1995). In our new model, all local activity is generated by statistical noise. We show that at least the low frequencies of EEG (up to ~20 Hz) can be explained by just the basic connectivity of the brain instead of local oscillators. This characteristic corresponds to a known feature of EEG where the lower frequencies (≤ 20 Hz) have strong global properties with a spatial distribution across the brain that depends on the frequency (Nunez, 2000; von Stein and Sarnthein, 2000; Nunez and Srinivasan, 2006; Srinivasan et al., 2013).

An account of whole brain dynamics would likely require at least mesoscopic models with macroscopic interactions (Nunez and Srinivasan, 2006; Breakspear, 2017; Glomb et al., 2022). In this paper, we show that the key feature of EEG, the alpha rhythm first discovered by Hans Berger, can be explained by a simple, linear model based on structural and functional connectivity instead of a local thalamic oscillator as suggested by Berger (Gloor, 1969).

1.2. Reducing the complexity of models based on structural and functional connectivity

Extensive literature focuses on using structural and functional connectivity measurements to inform global models (Glomb et al., 2022). However, the vast majority of the studies that exist in the literature also rely on embedded mesoscopic dynamics (e.g., Knock et al., 2009; Sanz-Leon et al., 2015), making it difficult to explore and make predictions from a global-only EEG model. We argue here that global-only models would be potentially useful for theory building to be related to cognitive phenomena in the field of Cognitive Neuroscience (Nunez et al., 2024). These models could also be helpful for clinicians given the importance of white matter in various disease states, development, cognitive decline and correlations with general cognitive abilities (Fields, 2008; Nunez et al., 2015; Ritchie et al., 2015; Simpson-Kent et al., 2020; Stammen et al., 2023).

Other previous work that models macroscopic brain dynamics assumes continuous features over space (e.g., Nunez, 2000; Raj et al., 2020), while our model instead assumes discrete spaces. These discrete spaces are separated into macroscopic brain areas for simplicity for modeling broad-level EEG features. Previous work also assumes relationships between local and macroscopic activity (e.g., Nunez, 2000; Raj et al., 2020), while our model assumes that all local activity is

noise (e.g., from a normal distribution reflecting *white noise*) and the dynamics are only driven by structural and functional connectivity. Furthermore, we assume distributions of axon delay times, derived below, instead of average delay times like previous global models (Nunez and Cutillo, 1995).

Our (linear) model of global EEG is purposefully simple. Papers that do not attempt linking mesoscopic models to macroscopic connections often present models with a large number of parameters (e.g., Nunez and Srinivasan, 2014). We have instead constrained our model to have a single functional parameter per connection and a structural distribution of delays per connection. We show that such simple statistical models can explain slow dynamics in EEG.

2. Materials and methods

2.1. Note on notation

For explicit communication we will use the following mathematical notation:

- $\mathbb{N}$ as a set of all natural numbers, $\mathbb{R}$ as a set of all real numbers.
- $f : X \mapsto Y$ to describe a function mapping elements from one set onto another, where
 - f is the function name,
 - X is the domain of the function f (the set of all possible inputs to the function),
 - Y is the codomain of the function f (the set of all possible outputs of the function).

For instance, the function $g(x, y) = x + y^2$, where $x \in \mathbb{R}$ and $y \in \mathbb{N}$, is characterized by $g : \mathbb{R} \times \mathbb{N} \mapsto \mathbb{R}$.

- $X_{\text{condition}}$ as an equivalent for $\{x \in X \mid \text{condition}(x)\}$, implying that $X_{\text{condition}}$ is a subset of X consisting of elements that satisfy a given condition. For instance, $\mathbb{R}_{\geq 0}$ represents all non-negative real numbers (rather than the commonly used $\mathbb{R}^+$ to avoid potential confusion with exponentiation notation).

2.2. Macroscopic model derivation

2.2.1. Deriving a macroscopic model from the mesoscale

Let R be a set of all regions in the brain defined on a mesoscale (Nunez and Srinivasan, 2006) and let T be the simulated time points. Note that R specifically denotes regions in the brain, distinct from $\mathbb{R}$, the set of real numbers. The modulations of the excitatory synaptic action $h : R \times T \mapsto \mathbb{R}_{\geq 0}$ and action potential fields about their background levels $g : R \times T \mapsto \mathbb{R}_{\geq 0}$ at some location $u \in R$ at time $t \in T$ follow a sigmoidal relationship, which is characteristic of population-level neural dynamics (Jirsa and Haken, 1997; Nunez, 2000). This relationship emerges naturally in macroscopic neural field models derived from quasi-microscopic descriptions of neural ensembles, where the conversion of synaptic inputs into action potentials exhibits a nonlinear sigmoidal form (Jirsa and Haken, 1997). Specifically, Eq. (1) reflects the balance between excitatory input, background excitability, and nonlinear inhibitory feedback, a formulation consistent with prior derivations of macroscopic neural field equations (Jirsa and Haken, 1997), where inhibitory feedback terms introduce cubic nonlinearity to stabilize the system:

$$\sum_{w \in R} \rho(u, w) g(u, t) = 2 \sum_{w \in R} \beta(u, w, t) h(u, t) - \alpha(u) h^3(u, t), \quad (1)$$

where

- $\rho : R \times R \mapsto \mathbb{N}$ represents the connectivity density, defined as the number of excitatory synapses per axon between regions;
- $\beta : R \times R \times T \mapsto \mathbb{R}_{\geq 0}$ represents background excitability between regions, or baseline excitatory drive that a region receives from another, perhaps due to neuromodulatory changes;

- $\alpha : R \mapsto \mathbb{R}_{\geq 0}$ reflects the strength of nonlinear inhibitory feedback, providing diminishing returns with increased excitatory activity.

In our work, we consider a linear approximation. Specifically, by setting $\alpha(u) \rightarrow 0$, we minimize the inhibitory synaptic action and treat it only as part of the background activity. The simplification leads to the following linear relationship between excitatory synaptic action h and action potentials g :

$$\sum_{w \in R} \rho(u, w) g(u, t) \approx 2 \sum_{w \in R} \beta(u, w, t) h(u, t). \quad (2)$$

2.2.2. Point-to-point connection model

The excitatory synaptic activity at one location (u) influences the excitatory activity of another (w), mediated by synaptic connections and signal propagation delays. Thus, besides local excitatory feedback and external input $z : R \times T \mapsto \mathbb{R}_{\geq 0}$, the excitatory synaptic action of region w at time t is also influenced by the excitatory input from region u to w with delay τ , such that $t - \tau \in T$:

$$\begin{aligned} h(w, t) = & z(w, t) \\ & + \rho(u, w) g(u, t - \tau) \\ & + \rho(w, w) g(w, t), \end{aligned} \quad (3)$$

Combining Eqs. (2) and (3), a linear relationship between excitatory synaptic activity in region w at times t and $t - \tau$ can be obtained. This follows from the assumption that excitatory synaptic action is proportional to action potentials via the background excitability:

$$\rho(u, w) g(u, t) = 2\beta(u, w, t) h(u, t). \quad (4)$$

Applying this to Eq. (3), we obtain:

$$\begin{aligned} h(w, t) = & z(w, t) \\ & + 2\beta(u, w, t - \tau) h(u, t - \tau) \\ & + 2\beta(w, w, t) h(w, t). \end{aligned} \quad (5)$$

In order to investigate what EEG power spectra can be accounted for by a truly *macroscopic* model, we assume *local* action potentials $g(u, t)$ and thus *local* synaptic action $h(w, t)$ are random. Given this assumption, we modify the equation to incorporate a stochastic term $\epsilon : R \times T \mapsto \mathbb{R}$. Therefore, the linear relationship between the excitatory synaptic activity in region w at times t and $t - \tau$ can be expressed as

$$\begin{aligned} h(w, t) = & z(w, t) \\ & + 2\beta(u, w, t - \tau) h(u, t - \tau) \\ & + \epsilon(w, t). \end{aligned} \quad (6)$$

This equation includes the combined effects of external input, delayed excitatory influences from another region, and the random variability of local synaptic action.

As mentioned above, the term 2β represents the baseline excitatory drive that a region receives from another. However, the connection strengths vary per pair of regions, and signal transmission times between them are not uniform, influenced by factors such as axon length and myelination. Therefore, 2β can be parsed into connection strengths $\gamma : R \times R \mapsto \mathbb{R}_{\geq 0}$ and the time lagged effects $\ell : R \times R \times \mathcal{P} \mapsto \mathbb{R}_{\geq 0}$ with $\mathcal{P} = \{t - \tau \in T \mid t, \tau \in T\}$, which quantify how the influence of region u on w changes with time delays:

$$2\beta(u, w, t - \tau) \rightarrow \ell(u, w, t - \tau) \gamma(u, w). \quad (7)$$

Substituting the parse into Eq. (6), we obtain

$$\begin{aligned} h(w, t) = & z(w, t) \\ & + \ell(u, w, t - \tau) \gamma(u, w) h(u, t - \tau) \\ & + \epsilon(w, t). \end{aligned} \quad (8)$$

2.2.3. All-to-point connection model

Extending Eq. (8) to include the influence of N regions across P discretized time lags, the following generalization is obtained:

$$\begin{aligned} h(w, t) = & z(w, t) \\ & + \sum_{n=1}^N \sum_{p=1}^P \ell(u_n, w, t - p\Delta t) \gamma(u_n, w) h(u_n, t - p\Delta t) \\ & + \epsilon(w, t). \end{aligned} \quad (9)$$

2.2.4. Macroscopic (all-to-all) model

Vector autoregressive (VAR) models are stochastic linear models that describe the relationship between multiple time-dependent variables. A VAR model states that the value of each variable at a given time is a linear combination of the values of all variables (sometimes including itself) at previous time points (Shumway and Stoffer, 2017).

When applying VAR models to study excitatory synaptic activity across different regions in the brain, each region's activity is treated as a stochastic variable. In this context, Eq. (9) represents the behavior of a single stochastic variable within a larger VAR framework. However, *vector* autoregressive models are usually written in terms of *all* stochastic variables to include the equations corresponding to all N regions. Then a VAR model of order P (i.e., VAR(P)) extends Eq. (9) to include all variables as follows:

$$\underline{h}(t) = \underline{z}(t) + \sum_{p=1}^P \left(\underline{\ell}(t - p\Delta t) \odot \underline{\gamma} \right) \underline{h}(t - p\Delta t) + \underline{\epsilon}(t). \quad (10)$$

In this form,

- $\odot$ denotes the *Hadamard product* (element-wise multiplication);
- $\underline{h} : T \mapsto \mathbb{R}_{\geq 0}^N$ is the vector of excitatory synaptic actions for all regions at time t ;
- $\underline{z} : T \mapsto \mathbb{R}_{\geq 0}^N$ is the vector of external inputs for all regions at time t ;
- $\underline{\ell} : \mathcal{P} \mapsto \mathbb{R}_{\geq 0}^{N \times N}$ is the matrix of lags, where each element $\ell(u, w, t - p\Delta t)$ represents the influence of region u at time $t - p\Delta t$ on region w at time t ;
- $\underline{\gamma} \in \mathbb{R}_{\geq 0}^{N \times N}$ is the matrix of connectivity weights, where each element $\gamma(u, w)$ represents the strength of the connection from region u to region w ;
- $\underline{\epsilon} : T \mapsto \mathbb{R}^N$ is the vector of random noise for all regions at time t .

In this particular study, we do not consider external stimuli, thus $\underline{z}(t) = \underline{0}$ for all $t \in T$. Hence, the model simplifies to

$$\underline{h}(t) = \sum_{p=1}^P \left(\underline{\ell}(t - p\Delta t) \odot \underline{\gamma} \right) \underline{h}(t - p\Delta t) + \underline{\epsilon}(t). \quad (11)$$

White matter pathways define structural connectivity between brain regions and shape global signal propagation. EEG properties have been shown to correlate with these connections (Valdés-Hernández et al., 2010). The time $\delta : R \times R \mapsto \mathbb{R}_{\geq 0}$ it takes for a signal to travel from region u to region w , thus the time lag between them, depends on the distance between these regions and the signal propagation speed:

$$\delta(u, w) = \frac{d(u, w)}{v(u, v)}, \quad (12)$$

where

- $d : R \times R \mapsto \mathbb{R}_{\geq 0}$ is the distance between two regions;
- $v : R \times R \mapsto \mathbb{R}_{\geq 0}$ is the signal propagation speed between two regions.

The relationship between axon diameter $\phi \in \mathbb{R}_{\geq 0}$ and conduction speed can be approximated with a linear function (Hursh, 1939):

$$v(u, v) \rightarrow v = k \phi, \quad (13)$$

where $k \in \mathbb{R}_{>0}$ is a constant factor estimated at 6 m per second per micron diameter (Drakesmith et al., 2019). We simplify the speed notation as we assume the axon diameters are independent from the brain regions.

Combining Eqs. (12) and (13), we obtain

$$\delta(u, w) = \frac{\lambda(u, w)}{\phi}, \quad \lambda(u, w) = \frac{d(u, w)}{k}. \quad (14)$$

Axon diameters between any two macroscopic regions vary, and we define the probability density function (PDF) of the distribution they follow as q_ϕ . Since $\lambda \in \mathbb{R}_{>0}$ in Eq. (14) is constant for a given pair of regions, the time lags are distributed according to the scaled inverse of the axon diameters with the PDF q_δ . To obtain the resulting distribution, we apply the change of variables method for probability density functions (PDFs), where the transformation between axon diameter and time lag induces a corresponding change in the distribution:

$$q_\delta(u, w, y) = \left| \frac{d}{dy} \left(\frac{\lambda(u, w)}{y} \right) \right| q_\phi \left(\frac{\lambda(u, w)}{y} \right), \quad (15)$$

where y represents a realization of the time lag between regions u and w .

As mentioned above, the time lagged effect ℓ quantifies how the influence of region u on w changes with time delays. This influence is inherently tied to the probability of a specific delay between regions occurring, which precisely is given by q_δ . Hence, we define

$$\ell(u, w, t - p\Delta t) = q_\delta(u, w, p\Delta t). \quad (16)$$

Combining Eqs. (11)–(16), we identify the following model parameters:

- The interregion distances d , which are fixed in an adult brain and cannot be changed based on the brain/cognitive state;
- The PDF of the axon diameter distribution q_ϕ , which we assume to be the same across individuals and brain/cognitive states;
- The connectivity weights γ , which can vary per individual and brain/cognitive state;
- The noise ϵ .

In the next subsection, we provide further details on the model components.

2.3. Model components

2.3.1. Brain regions

In an effort to avoid model complexity, we are intrinsically accounting for volume conduction effects by looking at large cortical surface areas. We focus on macroscale brain regions to capture the essential dynamics and interactions: the frontal (FL), parietal (PL), occipital (OL), and temporal (TL) lobes, as well as the thalamus (T). These lobes are chosen due to their significant roles in various cognitive functions, and the thalamus is included for its role in relaying neural signals.

Fig. 1 provides visualizations of these regions, each represented as a union of their subareas according to the Jülich-Brain Cytoarchitectonic Atlas (v3.0.3) (Amunts et al., 2022; Evans et al., 2012; Eickhoff et al., 2005). While most readers will be familiar with the locations of the cortical lobes, we include this figure to provide a clear visual reference for how these regions are defined within our framework. The lowest level subareas are listed in Appendix A.

2.3.2. Interregion distances

In this work, we assume that the distance between two regions is defined by the streamline length. Domhof et al. (2022) provided streamline lengths for 200 individuals obtained from diffusion-weighted MRI (dwMRI), detailing each brain region pair based on the Jülich-Brain parcellation (Amunts et al., 2020). For each pair, we take the 20% truncated mean across all individuals to define the distance.

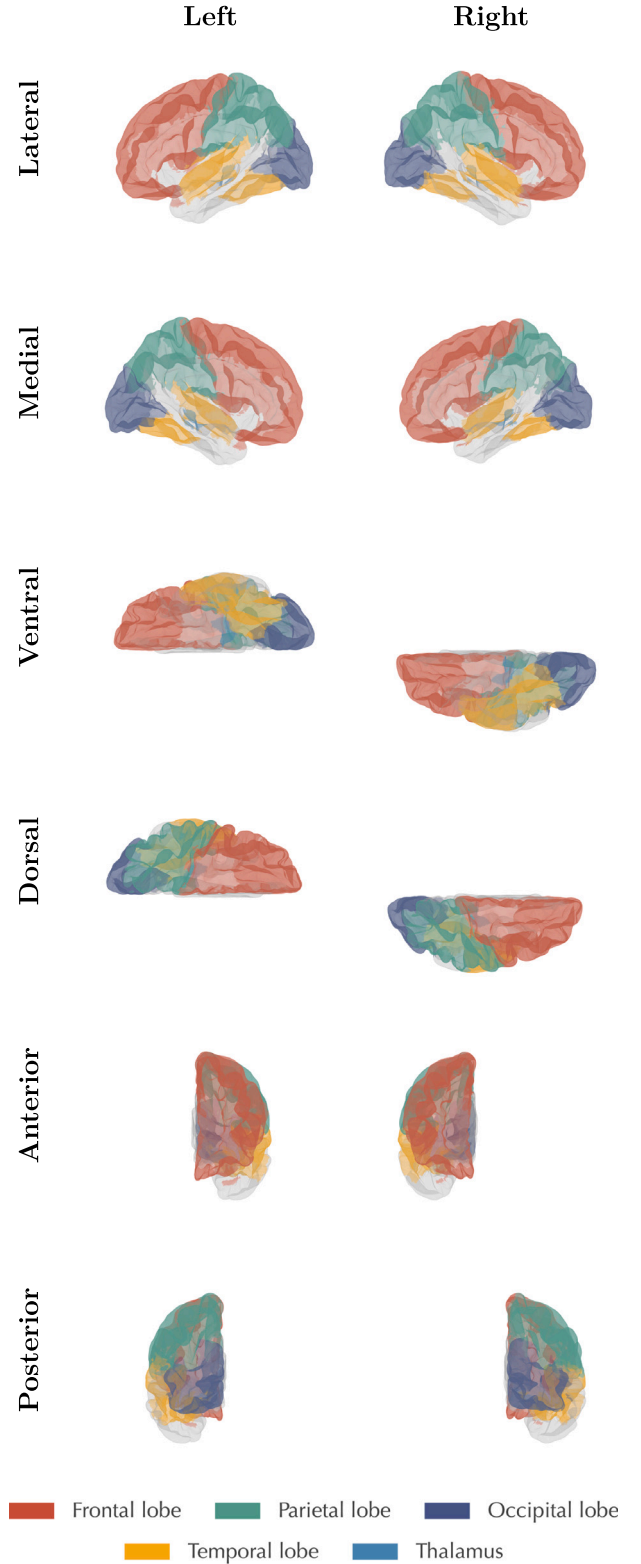


Fig. 1. A table of images visualizing the frontal, parietal, occipital, and temporal lobes, as well as the thalamus, each represented as a union of their subareas according to the Jülich-Brain Cytoarchitectonic Atlas (v3.0.3) (Amunts et al., 2022; Evans et al., 2012; Eickhoff et al., 2005). The regions are plotted on the FreeSurfer fsaverage pial surface mesh, displaying both left and right hemispheres (Dickscheid et al., 2023). For each hemisphere, the table shows lateral, medial, ventral, dorsal, anterior, and posterior views. The visualizations have been generated using the Nilearn package (Nilearn contributors, 2024).

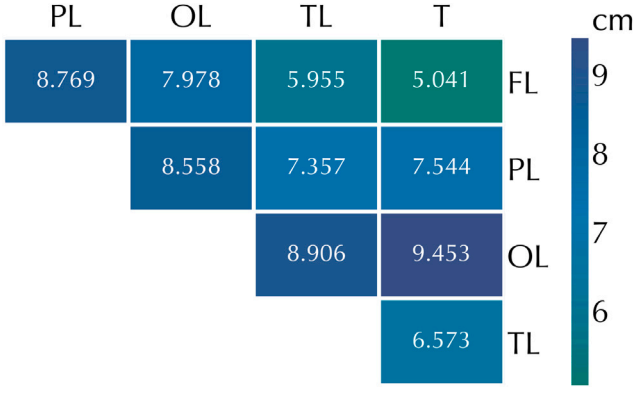
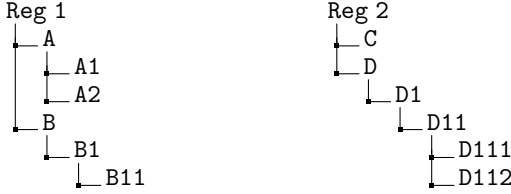


Fig. 2. Distances between selected brain regions in cm, calculated as the mean streamline lengths between subareas within each region (Domhof et al., 2022). The regions include the frontal, parietal, occipital, and temporal lobes, as well as the thalamus, based on the Jülich-Brain Cytoarchitectonic Atlas (v3.0.3) (Amunts et al., 2022; Evans et al., 2012; Eickhoff et al., 2005).

The Jülich-Brain atlas is hierarchical, and the brain areas chosen for the VAR simulation are not at the lowest level. So, the distance between two regions is calculated as the mean of all pairwise distances between the smaller subregions at the lowest level contained within the chosen larger regions.

Suppose we need to find the distance between two regions, Reg 1 and Reg 2, with the following hierarchical substructures:



The lowest level subregions of Reg 1 include A1, A2, and B11. The lowest level subregions of Reg 2 include C, D111, and D112. We define the distance between the two regions as

$$d(\text{Reg 1}, \text{Reg 2}) = \frac{\sum_{i,j \in \{A1, A2, B11\} \times \{C, D111, D112\}} d(i, j)}{|\{A1, A2, B11\}| \cdot |\{C, D111, D112\}|}. \quad (17)$$

The distance between the regions is computed as the mean of all pairwise distances between the lowest-level subregions, which is obtained by summing the distances over the Cartesian product of these subregions and dividing by the cardinality of this product.

The lowest level subregions of the selected brain regions are listed in Appendix A, as mentioned before. The resulting interregion distances are shown in Fig. 2.

2.3.3. Axon diameters

Axon diameters vary across streamline lengths, and the variability have been shown to follow Generalized Extreme Value (GEV) distribution (Sepehrband et al., 2016). Therefore, we assume that the distribution of axon diameters between two macroscopic regions follows the GEV as well. We denote the distribution as $\phi \sim \text{GEV}(\mu, \sigma, \xi)$ with the PDF q_ϕ :

$$q_\phi(x) = \frac{1}{\sigma} t(x)^{\xi+1} \exp(-t(x)), \quad (18)$$

$$t(x) = \left(1 + \xi \left(\frac{x - \mu}{\sigma}\right)\right)^{-\frac{1}{\xi}},$$

where $\mu \in \mathbb{R}$, $\sigma \in \mathbb{R}_{>0}$, and $\xi \in \mathbb{R} \setminus \{0\}$ are the location, scale, and shape parameters, respectively. We assume that the distribution of diameters is the same for all axons. The custom-selected values of these parameters are shown in Table 1.

Table 1

Summary of axon diameter distribution parameters (μ, σ, ξ) obtained from literature and custom estimates. The parameters from the studies pertain primarily to the corpus callosum, while the custom parameters are chosen to optimize the model.

Source	μ	σ	ξ
Sepehrband et al. (2016)	0.43	0.16	-0.21
Saeidi et al. (2023)	0.43	0.20	-0.30
Liewald et al. (2014)	0.53	0.46	-0.27
Custom	0.25	0.09	-0.25

Combining Eqs. (15) and (18), we can explicitly derive the scaled inverse Generalized Extreme Value distribution of lag effects ($\delta(u, w) \sim \text{InvGEV}(\mu, \sigma, \xi)$) with the PDF q_δ :

$$q_\delta(u, w, y) = \frac{\lambda(u, w)}{y^2} q_\phi\left(\frac{\lambda(u, w)}{y}\right) = \frac{\lambda(u, w)}{y^2} \frac{1}{\sigma} t\left(\frac{\lambda(u, w)}{y}\right)^{\xi+1} \exp\left(-t\left(\frac{\lambda(u, w)}{y}\right)\right), \quad (19)$$

$$t\left(\frac{\lambda(u, w)}{y}\right) = \left(1 + \xi \left(\frac{\lambda(u, w) - \mu y}{\sigma y}\right)\right)^{-\frac{1}{\xi}}.$$

2.3.4. Connectivity weights

The connectivity weights γ vary with cognitive states. We have custom-selected the parameters that align well with the EEG data at rest with eyes closed. In addition, we assume the values are symmetrical. The resulting values are shown in Fig. 3(a) and visualized in Fig. 4(a).

2.3.5. Noise with intrinsic volume conduction effects

Volume conduction effects are known to be pervasive in EEG recordings. Our model accounts for volume conduction by collapsing known connectivity over large areas of cortex. This is not an average measure of generated activity, but instead only an estimate of distributions over large areas. We would expect to observe this activity in real data based on certain assumptions in forwards models, such as the underlying activity coming from a large area on the cortical surface as opposed to single or double sources coming from deep in the cortex.

We use both white and pink noise to simulate local synaptic activity. Pink noise is thought to be more appropriate to account for volume conduction effects, which essentially acts as a low pass temporal filter due to fast oscillations typically being more localized (Pfurtscheller and Cooper, 1975; Nunez and Srinivasan, 2006; Nunez et al., 2019).

The pink noise, or $1/f$ noise, is employed as EEG signals often show power-law scaling behavior. Let $\eta_{\text{pink}} : R \times T \mapsto \mathbb{R}$ denote the pink noise for region w at time t . It is generated by transforming Gaussian white noise in the frequency domain to achieve a $1/f$ power spectral density (Timmer et al., 1995; Patzelt, 2023). The random noise ϵ is then defined as pink noise scaled by $s \in \mathbb{R}_{>0}$, which represents the standard deviation of the noise. This scaling adjusts the overall amplitude, effectively controlling the magnitude of power of the simulated EEG:

$$\epsilon(w, t) = s \eta_{\text{pink}}(w, t). \quad (20)$$

The white noise is characterized by a constant power spectral density, which can provide a baseline for understanding neural activity. Let $\eta_{\text{white}} : R \times T \mapsto \mathbb{R}$ denote the Gaussian white noise for region w at time t , where each sample is drawn from a normal distribution: $\eta_{\text{white}}(w, t) \sim \mathcal{N}(0, 1)$. The random noise ϵ is then similarly defined as:

$$\epsilon(w, t) = \left(s \sqrt{f_s}\right) \eta_{\text{white}}(w, t), \quad (21)$$

where $f_s \in \mathbb{N}$ is the sampling rate. Power is proportional to the square of the amplitude, hence we scale the noise by the square root of the sampling rate ratio. This ensures that the noise power remains consistent regardless of the sampling rate.

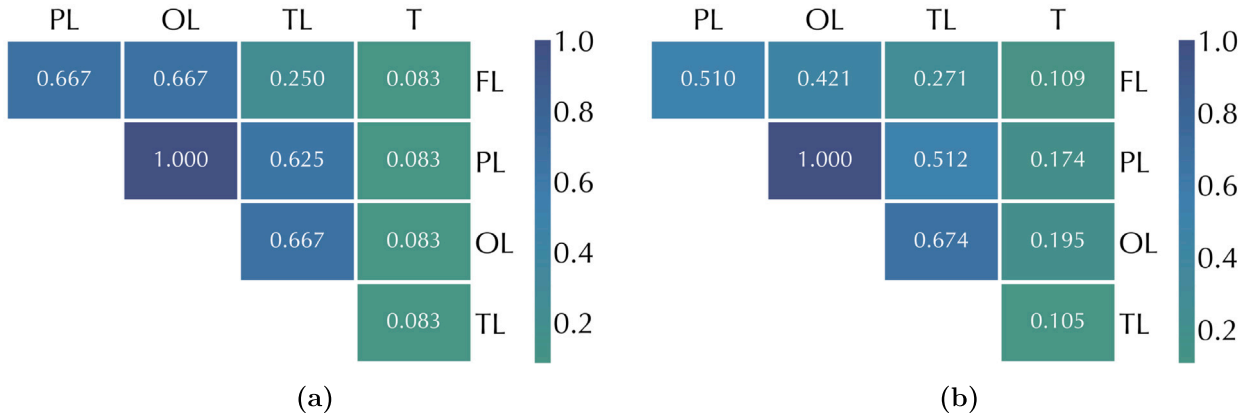


Fig. 3. The strength of neural connections among the frontal, parietal, occipital, and temporal lobes, as defined by the Jülich-Brain Cytoarchitectonic Atlas (v3.0.3) (Amunts et al., 2022; Evans et al., 2012; Eickhoff et al., 2005) for (a) custom-selected weights manually tuned to align with resting-state EEG features, (b) weights derived from resting-state fMRI functional connectivity using a power-law transformation with the exponent in power-law conversion from BOLD FC to weights $\chi = 1.5$ (Domhof et al., 2022).

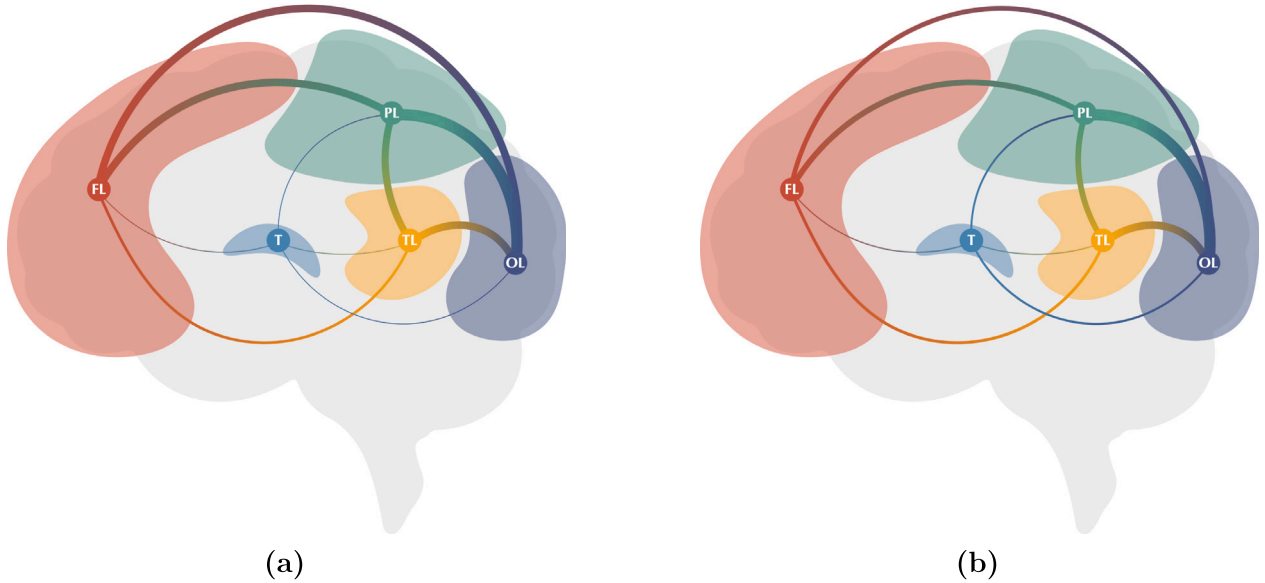


Fig. 4. The strength of neural connections among the frontal, parietal, occipital, and temporal lobes, as well as the thalamus visualized with line thickness for (a) custom-selected weights manually tuned to align with resting-state EEG features, (b) weights derived from resting-state fMRI functional connectivity using a power-law transformation with the exponent in power-law conversion from BOLD FC to weights $\chi = 1.5$ (Domhof et al., 2022).

2.3.6. Hypothetical literature-based parameters

In addition to the custom-selected values for the axon diameter distribution parameters and connectivity weights, we also utilize literature-based values that could hypothetically correspond to these parameters. We did not use empirically derived lag distributions from MEG/EEG as we felt this would result in circular logic when evaluating the model, though recent methods on estimating lags from MEG/EEG data are promising (e.g., Sorrentino et al., 2022).

The studies involving axon diameter distribution are primarily focused on the corpus callosum, with limited data available for other brain regions (Sepehrband et al., 2016; Saeidi et al., 2023; Liwald et al., 2014). The specificity and the variability in reported distributions suggest that these parameters may not accurately represent all brain areas. We have chosen to produce simulations with the parameters

derived from Sepehrband et al. (2016). A summary of both the custom-selected and literature-based parameter values is provided in Table 1, and the PDFs of the resulting distributions are shown in Fig. 5.

Additionally, to accommodate the literature-based axon diameter distribution, we redefine the interregion distances to employ the *relay station* approach. In this method, the thalamus serves as a relay station for signal transmission. The distance between two regions is thus calculated as the sum of the distances from each region to the thalamus.

When using the relay station approach, we derive the distribution of axon propagation speeds by drawing samples from the distributions of $\delta(u, \text{relay})$ and $\delta(w, \text{relay})$ and summing them to obtain the combined delay distribution. This combined distribution is then refined using Kernel Density Estimation (KDE), which provides a continuous estimate of the PDF for the combined delays.

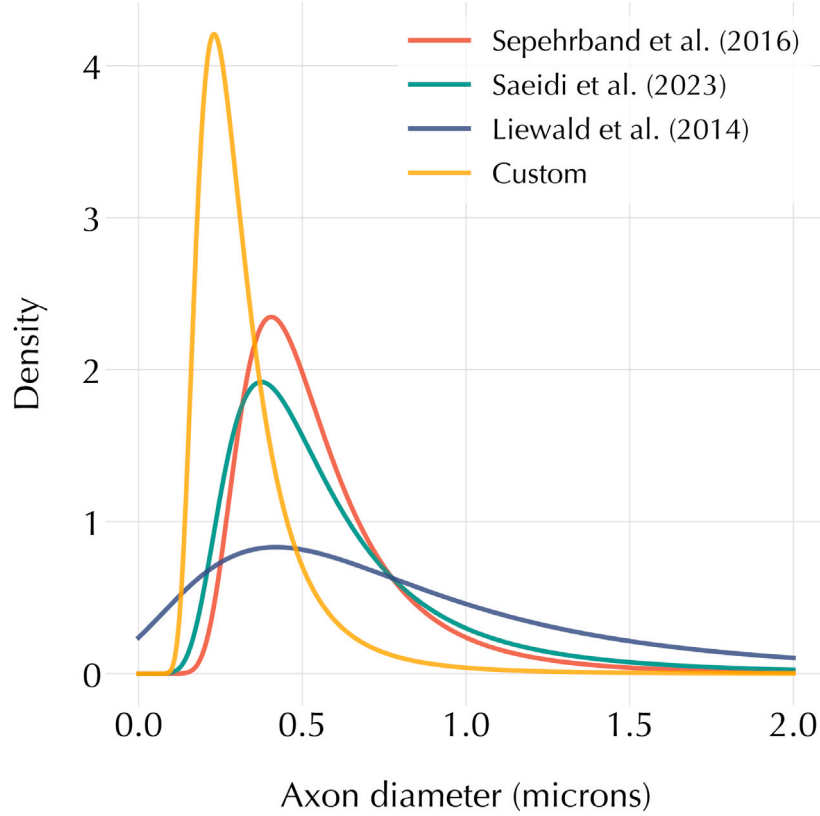


Fig. 5. Summary of probability density functions (PDFs) of axon diameter distributions based on parameters obtained from literature sources and custom estimates.

As a hypothetical measure of the connectivity between brain regions at rest, we use resting-state blood oxygen level dependent (BOLD) signals obtained via resting-state functional magnetic resonance imaging (rfMRI) (Hillman, 2014; Domhof et al., 2022). These signals capture the spontaneous fluctuations in brain activity and identify active neural networks in a resting brain.

Functional connectivity (FC) values, derived from Pearson correlation coefficients of the BOLD signal time series, are averaged across multiple sessions for each subject (Griffanti et al., 2014). Analogous to streamline lengths, we calculate FC between two brain regions from the Jülich-Brain atlas by taking the 20% truncated mean across individuals to define the connectivity.

The FC value, denoted as $b : R \times R \mapsto \mathbb{R}_{\geq 0}$, is computed as the mean of all pairwise FC values between the subregions that comprise the larger regions. To model the connection weight between two regions, we use the following equation:

$$\gamma(u, w) = \nu b(u, w)^\chi, \quad (22)$$

where $\nu \in \mathbb{R}_{>0}$ is a constant that scales the absolute value of the connection weights and $\chi \in \mathbb{R}_{>0}$ is an exponent that determines the nonlinearity of the relationship. The resulting connectivity weights are shown in Fig. 3(b) and visualized in Fig. 4(b).

Different functions were tested to map BOLD functional connectivity weights to VAR model connectivity weights. We found that the power law produced results consistent with EEG dynamics.

2.4. Simulations design

We perform four types of EEG simulations, two for each noise color (white and pink):

Table 2

Common parameters used for all simulations. $|T| = f_s \cdot (\text{Initialization time} + \text{Simulation time})$.

Parameter		Value
Simulation time	$ T $	500 s
Initialization time		10 s
Sample rate	f_s	1000 Hz
Number of lags	P	300
STD noise	s	1950 (white), 27 000 (pink)

- Custom-selected parameters (labeled as simulation #1);
- Literature-based parameters (labeled as simulation #2).

The relevant simulation parameters and their values are listed in Table 2.

In addition, we perform the simulations with the same parameters but for different sets of brain regions, in order to compare the results other global models with three regions, such as those of Nunez (2025).

- Custom-selected parameters for parietal, occipital, and temporal lobes (labeled as simulation #3);
- Literature-based parameters for parietal lobe, occipital lobe, and thalamus (labeled as simulation #4).

The results of these additional simulations are presented in Appendix B.

Before starting each simulation, we compute the weighted lag distribution effects $\underline{\Phi} : \mathcal{P} \mapsto \mathbb{R}_{\geq 0}^{N \times N}$, which are part of Eq. (11), such that

$$\underline{\Phi}(t - p\Delta t) = \underline{\mathcal{L}}(t - p\Delta t) \odot \underline{\gamma}. \quad (23)$$

In the VAR model, $\underline{\Phi}$ controls the dynamics of the system. Ensuring stability and stationarity of the system is essential to simulate bounded time series with invariant statistical properties over time.

To check stability and stationarity, we examine if the eigenvalues of the augmented matrix lie within the unit circle. We transform $\underline{\Phi}$ to have dimensions $N \times NP$. Then the augmented matrix $A \in \mathbb{R}_{\geq 0}^{NP \times NP}$ is defined as

$$A = \begin{pmatrix} \underline{\Phi} & \\ \mathbf{0}_{(P-1)N \times N} & \mathbf{I}_{(P-1)N} \end{pmatrix}. \quad (24)$$

If the matrix is not stationary, we scale the connectivity weights γ by 0.9:

$$\gamma \leftarrow 0.9 \gamma. \quad (25)$$

We repeat this process until reaching stationarity. Once we do, we proceed with simulating the EEG time series using the VAR model described in Eq. (11). We then filter the resulting excitatory synaptic actions using the highpass Butterworth filter with the passband frequency of 1 Hz (Virtanen et al., 2020). Next, we segment the simulated data $\underline{h}$ into 1 s intervals $\underline{h}'$ so its dimensions transform as follows:

$$|T| \times N \rightarrow f_s \times N \times \frac{|T|}{f_s}. \quad (26)$$

The segmentation is performed in order to improve the accuracy and reliability of spectral estimates. We then compute the power spectral density (PSD) of the signals corresponding to brain regions $G : R \times F \mapsto \mathbb{R}_{\geq 0}^N$ for frequencies of interest $F = \{0, \dots, |F|\}$ within Nyquist limits using the mean of power spectrum $\bar{G} : E \times R \times F \mapsto \mathbb{R}_{\geq 0}^{|F|}$ in each segment $e \in E = \{1, \dots, \frac{|T|}{f_s}\}$:

$$\begin{aligned} G(u, f) &= 2 \text{mean}_{e \in E} \bar{G}(e, u, f), \\ \bar{G}(e, u, f) &= \frac{|\mathcal{F}(h'(f, u, e))|^2}{f_s}, \end{aligned} \quad (27)$$

where $\mathcal{F}$ represents the Fast Fourier Transform (FFT).

2.5. Evaluation of alpha peaks

Statistical testing of systematic power in alpha vs. other ranges. We test the alpha peaks in power spectra to determine whether they are statistically significant. We analyze the segmented power values and apply statistical tests to assess the significance of the observed peaks compared to control frequency ranges.

However, in simulations with pink noise, a direct comparison of power values across frequency ranges is not appropriate due to the inherent bias from the $1/f$ characteristic. To correct for this, we detrend the power spectra by removing the $1/f$ trend while preserving total power:

$$\begin{aligned} \bar{G}(e, u, f) &\leftarrow \tilde{\bar{G}}(e, u, f) \sqrt{\frac{\sum_{f \in F} \bar{G}(e, u, f)^2}{\sum_{f \in F} \tilde{\bar{G}}(e, u, f)^2}}, \\ \tilde{\bar{G}}(e, u, f) &= \frac{\bar{G}(e, u, f)}{1/f}. \end{aligned} \quad (28)$$

To compare (detrended) power spectra for a region u , we compute mean power values across epochs for two categories: alpha $\bar{G}_{\text{alpha}}$ and control $\bar{G}_{\text{control}} : R \mapsto \mathbb{R}_{\geq 0}^{|E|}$. Specifically, we compute the mean power within the alpha frequency range $F_{\text{alpha}} = \{8, \dots, 13\}$ and the control frequency range $F_{\text{control}} = \{14, \dots, 20\}$ for each epoch:

$$\begin{aligned} \bar{G}_{\text{alpha}}(u) &= \{ \text{mean}_{f \in F_{\text{alpha}}} \bar{G}(e, u, f) \mid e \in E \}, \\ \bar{G}_{\text{control}}(u) &= \{ \text{mean}_{f \in F_{\text{control}}} \bar{G}(e, u, f) \mid e \in E \}. \end{aligned} \quad (29)$$

To determine whether the alpha power peaks are statistically significant, we compare $\bar{G}_{\text{alpha}}$ and $\bar{G}_{\text{control}}$ using paired statistical hypothesis testing. Specifically, we test whether the mean power in the alpha range is significantly greater than in the control range across epochs.

Let the difference in power $\Delta \bar{G}$ for an epoch e be defined as follows:

$$\Delta \bar{G}(e) = \bar{G}_{\text{alpha}}(e) - \bar{G}_{\text{control}}(e). \quad (30)$$

To determine whether the observed differences indicate a systematic increase in alpha power, we perform a one-sided hypothesis test:

H_0 : $\text{mct}(\Delta \bar{G}) \leq 0$ (there is no significant increase in alpha power compared to the control power),

H_1 : $\text{mct}(\Delta \bar{G}) > 0$ (alpha power is significantly greater than control),

where mct is a measure of central tendency.

The choice of statistical test depends on whether the differences in power are normally distributed. To test normality, we use the *Shapiro–Wilk test* (SW) on $\Delta \bar{G}$.

- If they are normally distributed (SW p-value ≥ 0.05), we apply a *paired t-test*, where $\text{mct} := \text{mean}$;
- if they are not normally distributed (SW p-value < 0.05), we apply the *Wilcoxon signed-rank test*, a non-parametric alternative, where $\text{mct} := \text{median}$.

The corresponding one-tailed p -value determines whether we reject H_0 in favor of H_1 :

- p-value < 0.05 : we reject H_0 ; alpha power is significantly greater than the control power;
- p-value ≥ 0.05 : we fail to reject H_0 ; there is no sufficient evidence to conclude that alpha power is significantly greater.

The statistical tests are conducted on each individual simulation of a given simulation type, and we report the mean p-value across the $S = 100$ simulations of that type.

Testing of alpha peaks with FOOOF. In addition to the statistical tests of alpha power, we evaluate the presence of distinct peaks in the alpha range using the *fitting oscillations & one over f* (FOOOF) method (Donoghue et al., 2020). For each simulation type, we use the FOOOF algorithm to parameterize the power spectrum and identify the presence of peaks, independent of the $1/f$ noise. The FOOOF fitting is constrained with peak width limits of 2–8 Hz and a maximum of 10 peaks per spectrum, using a fixed aperiodic mode.

Specifically, for each simulation type, we perform S simulations. For each simulation and brain region, we compute a binary indicator $\mathbb{1}_b : \{1, \dots, S\} \times F \times R \mapsto \{0, 1\}$ that identifies whether a peak is present at a given frequency (1) or not (0). We then compute two metrics for each region u in a simulation type:

1. % in F_{alpha} – the percentage of simulations that exhibit at least one peak in the alpha range F_{alpha} ,
2. % of F_{alpha} – the percentage of all detected peaks (across the entire frequency spectrum) that are located within the alpha range.

2.6. Evaluation with coherence

For each brain region pair on the scalp, we relate their simulated signals. Let $R' = R \setminus \{T\}$ (the thalamus is excluded as it does not have a direct surface representation in the EEG data). We then compute the relations by computing pairwise coherence $C : R' \times R' \times F \mapsto [0, 1]$ between regions:

$$C(u, w, f) = \frac{|G(u, w, f)|^2}{G(u, f) G(w, f)}, \quad (31)$$

where $G(u, w, f)$ is the cross-spectral density (CDF) between $G(u, f)$ and $G(w, f)$.

3. Results

3.1. Simulations

Figs. 10 and 7 include the results of the single runs of each type of simulations: Fig. 10 shows the lag effects as well as the power plots with both white and pink noise for simulation #1, i.e., the simulations with custom-selected parameters. Similarly, Fig. 7 shows the equivalent plots for simulation #2, i.e., with literature-based parameters. Appendix C includes the power plots of individual brain regions.

Figs. 6(a) and 7(a) illustrate the weighted effects of propagation delays (Φ) for the simulations. The delays in simulation #1 are generally shorter and have a larger right tail. In both Figures, the pair with the largest relative lag effect is PL-OL. Additionally, all connections with the thalamus (T) have the smaller delay effects than pairings with other regions.

Figs. 6(b) and 7(b) show the power spectrum (G) for the simulations performed with white noise. Both results include peaks in the alpha band (8–13 Hz), with the occipital (OL) and parietal (PL) lobes yielding the highest power. Moreover, both Figures exhibit high power in the delta band (under 4 Hz). In beta (13–20 Hz) and gamma (20–50 Hz) bands, the power resembles white noise in all regions except the occipital lobe in simulation #1 which is slightly elevated in low gamma.

Figs. 6(c) and 7(c) show the power spectrum (G) for the simulations performed with pink noise. The peaks are similar to those in white noise simulations. However, the power in delta is much higher. In addition, the power in beta and gamma decreases asymptotically.

Appendix D shows the effect of the choice of the axon distribution parameters on the simulation results. Appendix E includes the power plots for S simulations of each simulation type.

3.2. Coherence

Fig. 8 includes the pairwise coherence values (C) for the simulated EEG data shown in Figs. 10 and 7. Due to the absence of the volume conduction effect in the simulations, the coherence values for all simulated data are lower than those of the real EEG data (Nunez and Srinivasan, 2006). Attention should be given to the peaks in coherence and their locations.

For all pairs of regions, simulated coherence is high in low frequencies (delta band) and is decreasing exponentially. This feature is missing in the real EEG data (Nunez and Srinivasan, 2006). Additionally, all pairs except FL-TL have a coherence peak in the alpha band, while the pair FL-TL shows peaks in the beta band. In the real EEG, the alpha peak in coherence is expected to appear between all pairs of brain regions with varying magnitudes (e.g., Figure 9.17 in Nunez and Srinivasan, 2006). The most prominent peak for the simulated EEG is in PL-OL.

3.3. EEG features

Fig. 9 presents the results of statistical tests across S simulations assessing whether alpha power is significantly greater than control power across brain regions, confirming the presence of characteristic alpha peaks. However, there is insufficient evidence to support the presence of an expected alpha peak in the occipital lobe (OL) for simulation #2 under both white and pink noise conditions. Moreover, while the p-value for the parietal lobe (PL) is below 0.05, it is relatively high. We attribute these results for simulation #2 to the fact that the peak in those simulations appears to be slightly shifted towards lower frequencies.

The proportion of simulations in which at least one alpha peak is detected (% in F_{α}), the percentage of all detected peaks that are located within the alpha band (% of F_{α}), and the corresponding peak plots for each simulation type are provided in Appendix F. Consistent with the results of the statistical tests, the proportion of peaks in

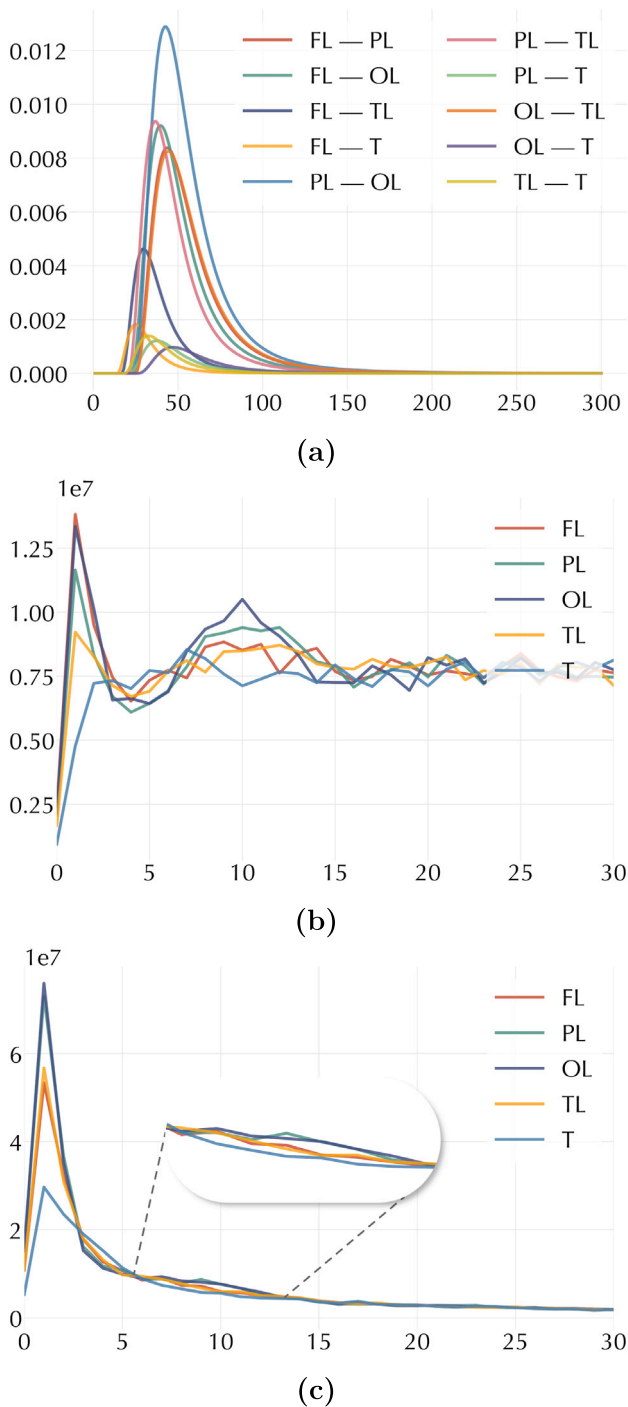


Fig. 6. Plots of propagation delays effects scaled by connectivity weights and the power spectra of simulated signals for simulation #1: (a) weighted lag effects for each lag (ms), (b) power at each frequency (Hz) for white noise, (c) power at each frequency (Hz) for pink noise.

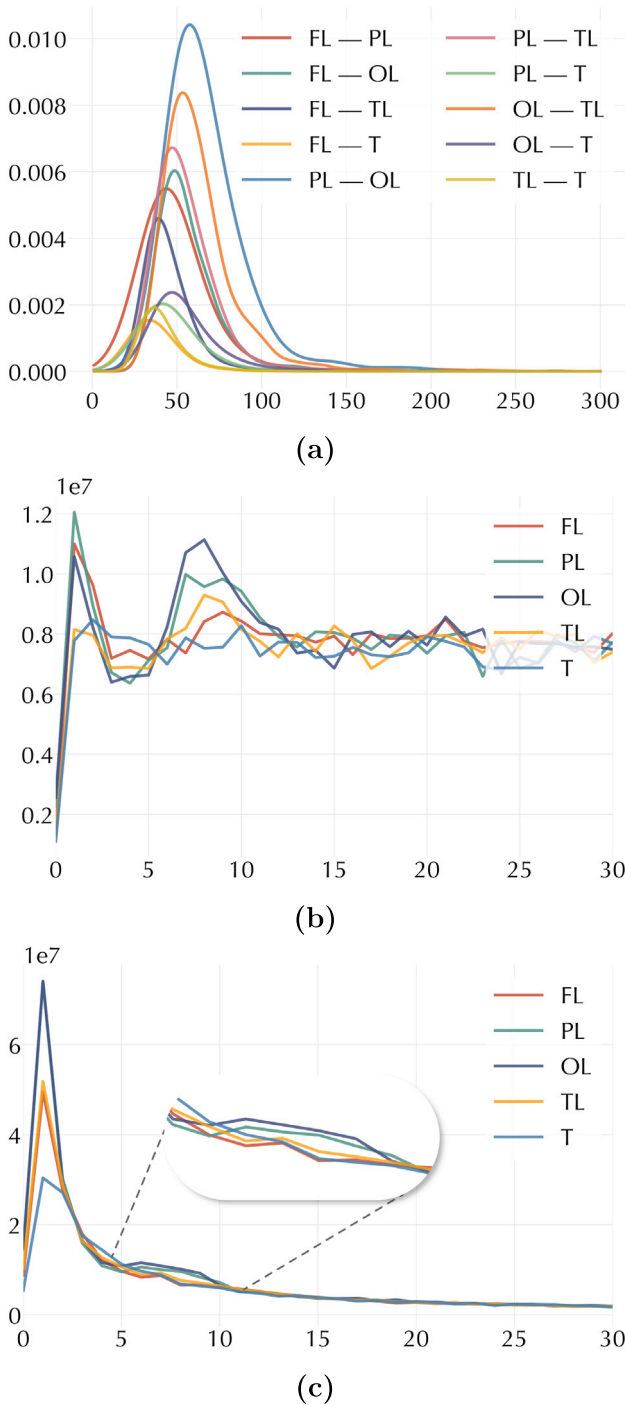


Fig. 7. Plots of propagation delays effects scaled by connectivity weights and the power spectra of simulated signals for simulation #2: (a) weighted lag effects for each lag (ms), (b) power at each frequency (Hz) for white noise, (c) power at each frequency (Hz) for pink noise.

the alpha band and their density is the largest for PL and OL in all simulations.

Table 3, informed by these results, presents a summary of the ability of the simulated models to reproduce power- and coherence-related characteristics observed in the real EEG data. Simulation #1 with both pink and white noise generates alpha power in the occipital and parietal lobes. Simulation #2 does not clearly generate alpha power in the occipital lobe. All simulations generate expected coherence alpha peaks, as can be seen in Fig. 8. However, the simulated data exhibit high coherence in low frequencies, a feature absent in real EEG data.

4. Discussion

In this work, we show that frequency band features of scalp-recorded EEG at rest can be described by a purely macroscopic model of connection delay distributions. The model is simple in that it relies only on linear parameter relationships in a vector autoregressive (VAR) framework, and the coefficients between two nodes can be explained entirely by four parameters: a distribution of propagation delays (defined by three parameters) and a connectivity weight. Specifically, we have shown that resting eyes-closed alpha rhythm power and coherence can be explained by frontal lobe to occipital lobe connections with anatomy-derived delay distributions. This conclusion is especially true when treating thalamic connections as a *relay station* that extends the timing of these connections. Connection strengths have been derived from fMRI BOLD activity using an exponential conversion function. This function has been found to maximize correspondence between the simulated and real EEG measures. We have not found information about this conversion function in the literature, but it is a prediction that can be tested in future joint fMRI-EEG data collections.

4.1. In defense of simple models

Our work is a purposeful oversimplification of the complexities of brain electrophysiology. The goal of this work is not to capture *all* features of scalp-recorded EEG with a simple model. We ignore local activity completely as well as some theorized macroscale activity, such as those resulting from short-range U-fibers (Nunez, 2025). This work also ignores the interactions between local and macroscale scales (Nunez, 2000). Similar models could have used different (e.g., finer) brain parcellations instead of the cortical lobes themselves. Instead, we show how a very simple macroscale model can capture many of the features of resting EEG. We feel this modeling work is the first step towards the ability to *fit* models with a low number of estimated parameters to real EEG data to discover individual- and cognitive state-differences in experimental data.

There is also a need for simple models in the field of model-based Cognitive Neuroscience (Schubert et al., 2019; Nunez et al., 2024; Forstmann and Turner, 2024). EEG models with low numbers of parameters could be used to help inform individual differences and experiment-level differences in cognition, especially when combined with *cognitive* models that describe psychological processes. These psychological processes can then be used as *measurement models* in order to measure, for instance, individual differences in the psychological construct of evidence accumulation during decision-making (e.g., Nunez et al., 2025). Combining simple computational models of electrophysiological dynamics with neurocognitive models could lead to a richer understanding of human neurocognition as reflected by real EEG and behavioral data in human participants.

4.2. Limitations

A simple linear account that relies only on macroscopic brain behavior clearly misses some features of resting EEG. A drastic feature of the

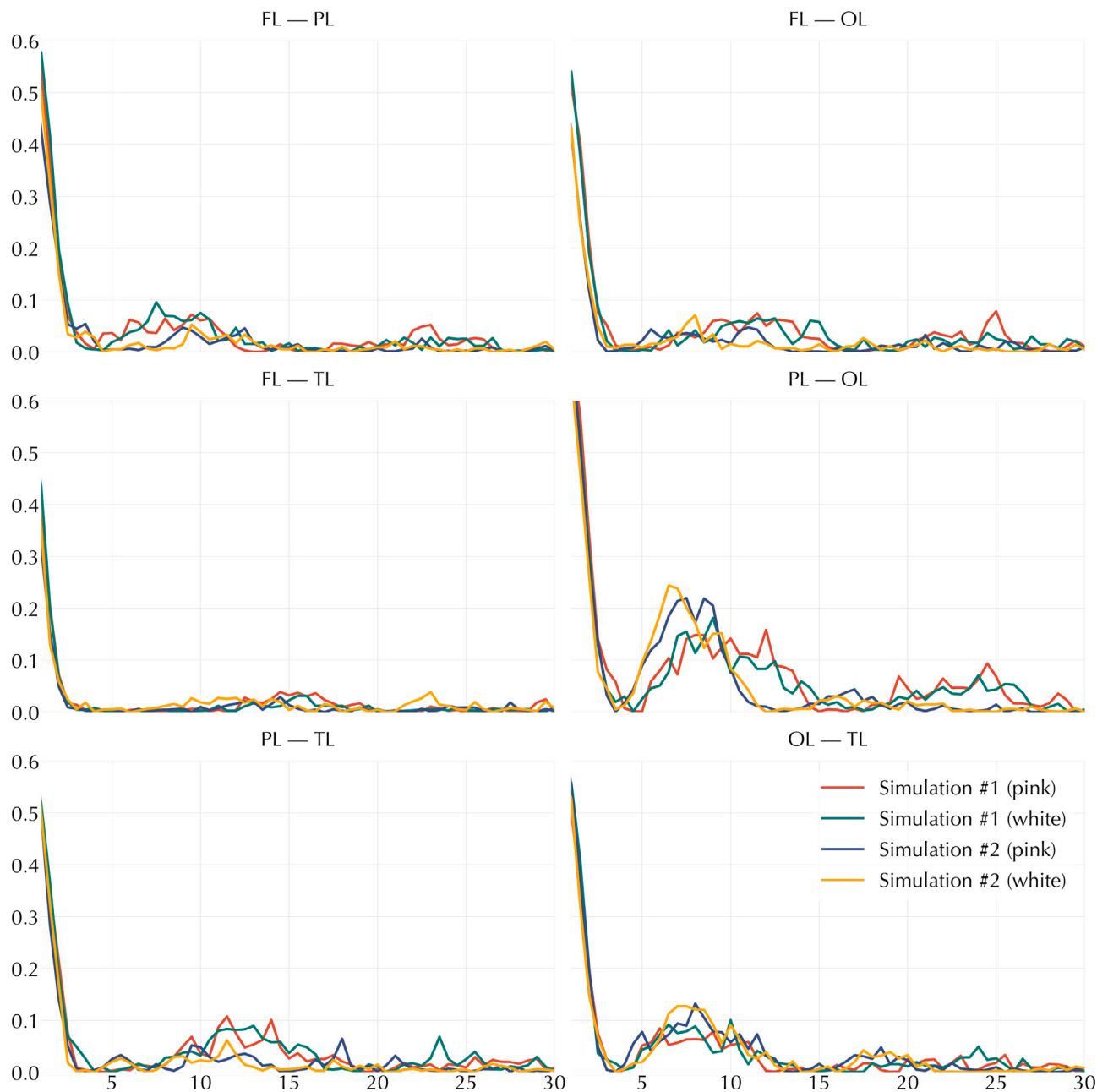


Fig. 8. Pairwise coherence between brain regions for the simulations.

Table 3				
The summary of the abilities of the simulated macroscopic models in replicating classical properties of scalp EEG.				
Prominent feature	Simulation #1 (white) Fig. 6(b)	Simulation #1 (pink) Fig. 6(c)	Simulation #2 (white) Fig. 7(b)	Simulation #2 (pink) Fig. 7(c)
Power alpha peak in PL	✓	✓	✓	✓
Power alpha peak in OL	✓	✓	?	?
Coherence alpha peak in FL-OL	✓	✓	✓	✓
Coherence alpha peak in PL-OL	✓	✓	✓	✓
Coherence alpha peak in OL-TL	✓	✓	✓	✓
No high coherence at low frequencies	×	×	×	×

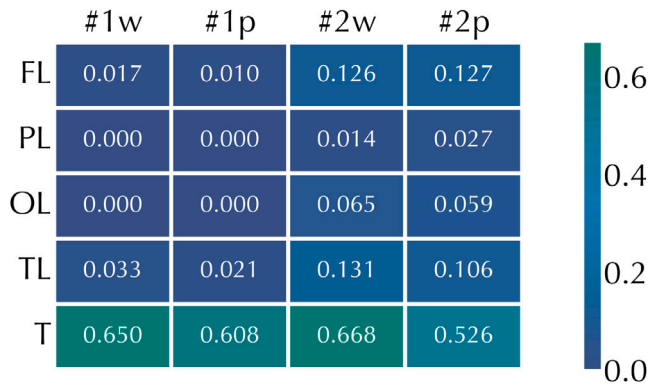


Fig. 9. Results of the statistical tests assessing whether alpha power is significantly greater than control power. For each brain region, the corresponding mean p -value across S simulations rounded to three decimal places is presented. Simulation types are abbreviated as #1w (simulation #1 with white noise), #1p (simulation #1 with pink noise), etc.

simulated data is the high coherence in lower frequencies, < 5 Hz, while real EEG records have no such coherence (e.g., compared to Figure 9–18 by Nunez and Srinivasan (2006)). We found this surprising given the analytic solutions of Nunez and Srinivasan (2014), who found that various idealized global models predict damping of low frequencies for certain excitability levels. Though this could be due to fundamental differences between the models: our model is limited only to interregion macroscopic connections, while the models by Nunez and Srinivasan (2014) are also driven by broad unspecified global connectivity (as could be instantiated with U-fibers, see Nunez (2025)). Another feature of some of the simulations is the power and coherence of harmonics, e.g., power at both 9 and 18 Hz in some simulations, though this is unsurprising given predictions by Nunez and Srinivasan (2014). We report realistic simulations that did not have these harmonic effects.

The presence of high frequency activity > 20 Hz in our simulations that assumed local white noise was also surprising and not reflective of real EEG data (e.g., see log power plots of Whitham et al., 2007). We aimed to fix this with pink noise to approximate volume conduction effects, but this approach resulted in high power in the low frequencies. We suspect that there is a type of noise that results in an optimal trade-off between realistic power in high and low frequencies. However, we felt this would be an overly restrictive assumption in the model that was not theoretically well justified. It is known that volume conduction reduces high frequencies, and this reduction occurs due to a property of the brain that low frequencies tend to be more spatially distributed (Pfurtscheller and Cooper, 1975; Nunez and Srinivasan, 2006; Nunez et al., 2019). We know that a more complex model of global activity would result in more realistic power spectra (e.g., Nunez and Srinivasan, 2014; Raj et al., 2020). However, this approach would result in a less simple model. In future work, we aim to develop and explore simple models that result in more realistic EEG power spectra.

Finally, we hypothesized that recurrent connections operationalized by a weighted moving average process would result in a higher correspondence between the simulations and real data. However, this was not the case. It is possible that non-linear connections could change these features in future modeling work. Extensions of our macroscopic model to include mesoscopic effects would likely fix the missing description of some effects, but it is unclear if such extensions would be necessary to capture slow wave EEG dynamics.

4.3. Possible uses of simple macroscopic models

The models presented in this manuscript could be used by researchers to better quantify theories about how structural brain differences can give rise to differences in global features of EEG. For instance, it has been shown that frontal and parietal/occipital alpha power and coherence changes as children age, and this is thought to be associated with large white matter volume increases and/or increased myelination (Thatcher et al., 1987; Srinivasan, 1999). In our model, a larger number of axons can be simulated by larger connectivity weights γ in older individuals in all cognitive states, while increased myelination could be simulated with different axon diameter distributions q_ϕ across age groups (Lynn et al., 2021). During childhood development, specific alpha peaks are also thought to increase in frequency (Lindsley, 1939; Marshall et al., 2002) in addition to other frequency components being more prevalent. These phenomena are not likely to be explained in our model by simulating changing interregion distances d over development. The brain grows over development, interregional distances become larger, and the peak frequency should become slower. However, other physiological changes modeled by changing connectivity weights γ and axon diameter distributions q_ϕ may show that the peak frequency increases.

The model can also be informed by measures from MRI and other imaging techniques, as we have shown in this paper. Simulated data from our model that is informed by imaging data can yield EEG predictions for differences between healthy and clinical populations, such as psychiatric and neurological pathologies. These predictions can also be individualized with EEG simulations informed by patient imaging. For instance, there has been some success in showing broad EEG differences in stroke patients compared to health controls (e.g., see Wu et al., 2016; Singh et al., 2024). Simulating individual patients' data in simple models may explain some of the empirical findings and help develop hypotheses about patients' EEG. The models may also be extended to clinical domains with electrophysiological pathologies in local areas spreading globally, such as in epilepsy patients. However, the clinical usefulness of these simple models of EEG and possible extensions remain to be tested and developed.

CRedit authorship contribution statement

Mariia Steeghs-Turchina: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Ramesh Srinivasan:** Writing – review & editing, Methodology, Conceptualization. **Paul L. Nunez:** Writing – review & editing, Methodology. **Michael D. Nunez:** Writing – review & editing, Writing – original draft, Supervision, Software, Project administration, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

I have nothing to declare.

Appendix A. Regions subareas

Table 4 lists subareas of frontal, parietal, occipital, and temporal lobes, as well as the thalamus at lowest level of the hierarchical organization, according to the Jülich-Brain Cytoarchitectonic Atlas (v3.0.3) (Amunts et al., 2022; Evans et al., 2012; Eickhoff et al., 2005). The areas represent the smallest distinct anatomical units identified in the atlas.

Appendix B. Results for three regions

See Figs. 10 and 11.

Table 4

List of all subareas at the lowest level of the frontal, parietal, occipital, and temporal lobes, as well as the thalamus, according to the Jülich-Brain Cytoarchitectonic Atlas (v3.0.3) (Amunts et al., 2022; Evans et al., 2012; Eickhoff et al., 2005).

Frontal lobe	Parietal lobe	Occipital lobe	Temporal lobe	Thalamus
Area IFJ1 (IFS, PreCS)	Area 5L (SPL)	Area hOc2 (V2, 18)	Area STS2 (STS)	CGM (Metathalamus)
Area IFS4 (IFS)	Area 7M (SPL)	Area hOc1 (V1, 17, CalcS)	Area STS1 (STS)	CGL (Metathalamus)
Area IFS2 (IFS)	Area 7PC (SPL)	Area hOc6 (POS)	Area TE 3 (STG)	
Area IFS1 (IFS)	Area 5M (SPL)	Area hOc4d (Cuneus)	Area TE 2.2 (STG)	
Area IFS3 (IFS)	Area 7P (SPL)	Area hOc3d (Cuneus)	Area TE 2.1 (STG)	
Area IFJ2 (IFS, PreCS)	Area 5Ci (SPL)	Area hOc3v (LingG)	Area TI (STG)	
Area MFG2 (MFG)	Area 7A (SPL)	Area hOc4v (LingG)	Area TeI (STG)	
Area 8v2 (MFG)	Area OP3 (POperc)	Area hOc4lp (LOC)	Area TE 1.2 (HESCHL)	
Area 8v1 (MFG)	Area OP4 (POperc)	Area hOc5 (LOC)	Area TE 1.1 (HESCHL)	
Area MFG1 (MFG)	Area OP2 (POperc)	Area hOc4la (LOC)	Area TE 1.0 (HESCHL)	
Area 44 (IFG)	Area OP1 (POperc)		Area Ph1 (PhG)	
Area 45 (IFG)	Area 3b (PostCG)		Area Ph2 (PhG)	
Area 6d2 (PreCG)	Area 1 (PostCG)		Area Ph3 (PhG)	
Area 6d1 (PreCG)	Area 2 (PostCS)		Area FG2 (FusG)	
Frontal-I (GapMap)	Area 3a (PostCG)		Area FG3 (FusG)	
Frontal-II (GapMap)	Area PF (IPL)		Area FG1 (FusG)	
Area 6ma (preSMA, mesial SFG)	Area PFcm (IPL)		Area FG4 (FusG)	
Area Fp1 (FPole)	Area PGa (IPL)		Area CoS1 (CoS)	
Area Fp2 (FPole)	Area PFt (IPL)		Area TPJ (STG/SMG)	
Area SFS2 (SFS)	Area PFm (IPL)			
Area 6d3 (SFS)	Area PGp (IPL)			
Area SFS1 (SFS)	Area PFop (IPL)			
Area 6mp (SMA, mesial SFG)	Area hPO1 (POS)			
Area 4p (PreCG)	Area hIP1 (IPS)			
Area 4a (PreCG)	Area hIP7 (IPS)			
Area Fo1 (OFC)	Area hIP3 (IPS)			
Area Fo3 (OFC)	Area hIP2 (IPS)			
Area Fo2 (OFC)	Area hIP4 (IPS)			
Area Op7 (Frontal Operculum)	Area hIP5 (IPS)			
Area Op8 (Frontal Operculum)	Area hIP6 (IPS)			
Area Op5 (Frontal Operculum)	Area hIP8 (IPS)			
Area Op6 (Frontal Operculum)				
Area Op9 (Frontal Operculum)				
Area Fo6 (OFC)				
Area Fo5 (OFC)				
Area Fo7 (OFC)				
Area Fo4 (OFC)				
Area 8d2 (SFG)				
Area 8d1 (SFG)				

Appendix C. Powers per brain region

Fig. 12 shows the power plots produced by the single simulations.

Appendix D. Lag distributions and results

Fig. 13 provides a collection of weighted lag effects with varying parameter values (μ, σ, ξ) for the hypothetical distributions of the axon diameters, modeled using the Generalized Extreme Value (GEV) distribution. Fig. 14 shows the power plots of the simulations with corresponding weighted lag effects.

As the location parameter μ increases, the so-called Inverse Generalized Extreme Value distribution of the lag effects shifts to the left and becomes narrower (see the rows in Fig. 13). As the scale parameter σ increases, the distribution shifts to the left (see columns of three adjacent plots in Fig. 13). Finally, as the shape parameter ξ increases, the distribution becomes more narrow (see columns with every third plot in Fig. 13). As evident from Fig. 14, distributions shifted to the left have peaks at higher frequencies. Narrower distributions lead to higher power, and when this power reaches a sufficiently high level, harmonics of the peaks become visible.

Appendix E. Power plots for 100 simulations

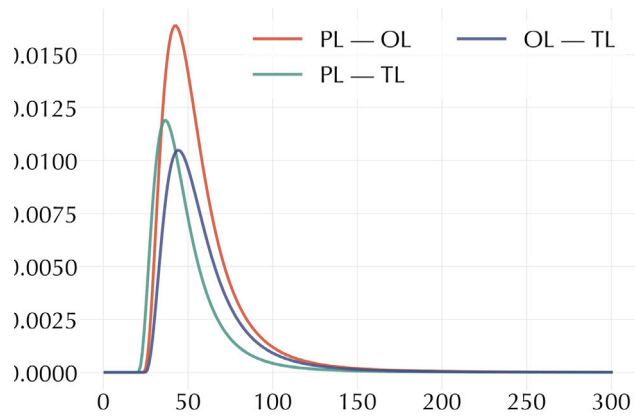
Figs. 15 and 16 show the mean power plots with standard deviations produced by 100 simulations.

Appendix F. FOOOF peaks

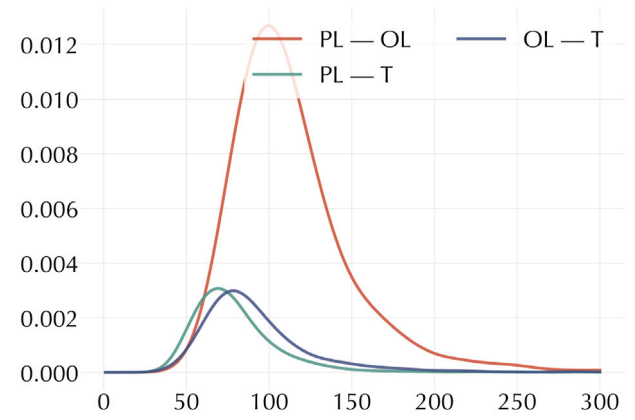
Figs. 17 and 18 show the peak presence plots across 100 simulations. Figs. 19 and 20 show the evaluation of the presence of distinct alpha peaks across S simulations: Fig. 19 illustrates the proportion of simulations in which at least one alpha peak is detected (% in F_{α}); Fig. 20 shows the relative density of alpha peaks, i.e., the percentage of all detected peaks that are located within the alpha band (% of F_{α}).

Data availability

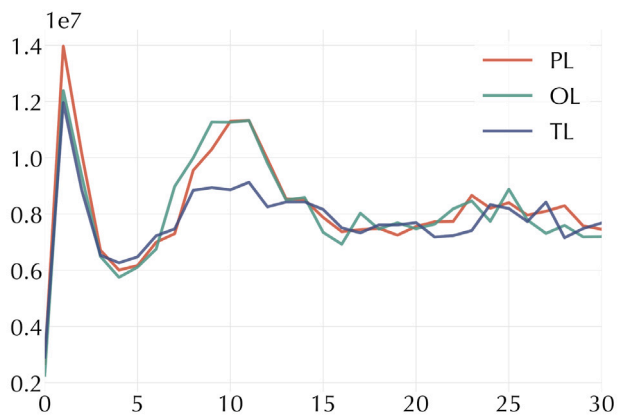
The code used to perform the simulations can be found at <https://github.com/thatmariia/macro-eeg-model> (v0.3.0) (Steeghs-Turchina et al., 2025). The data reported in this paper is archived at <https://doi.org/10.5281/zenodo.13618554> (Steeghs-Turchina et al., 2024).



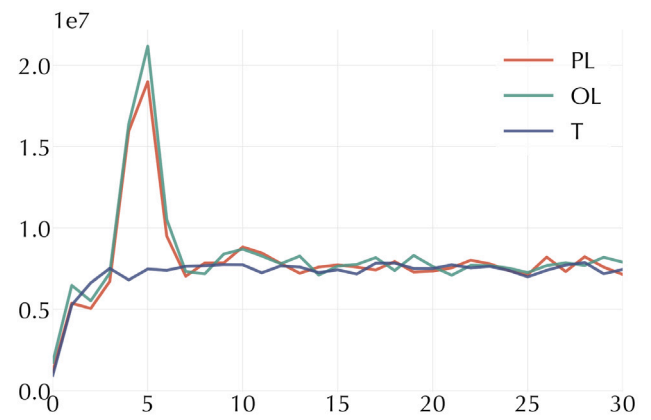
(a)



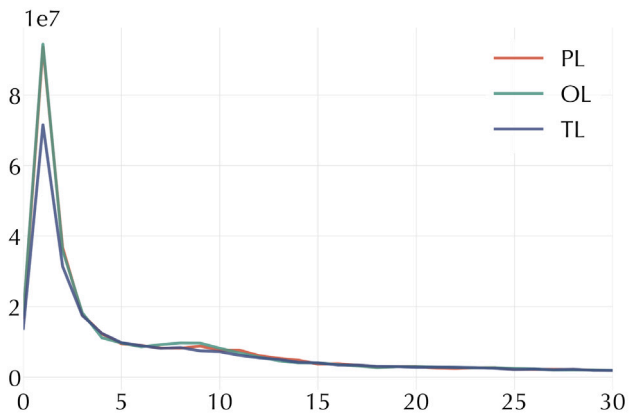
(a)



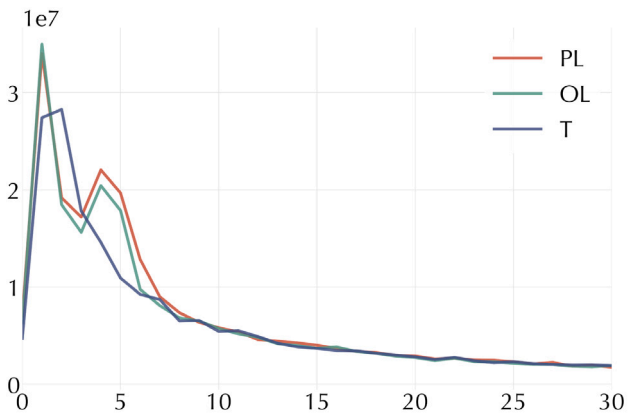
(b)



(b)



(c)



(c)

Fig. 10. Plots of propagation delays effects scaled by connectivity weights and the power spectra of simulated signals for simulation #3: (a) weighted lag effects for each lag (ms), (b) power at each frequency (Hz) for white noise, (c) power at each frequency (Hz) for pink noise.

Fig. 11. Plots of propagation delays effects scaled by connectivity weights and the power spectra of simulated signals for simulation #4: (a) weighted lag effects for each lag (ms), (b) power at each frequency (Hz) for white noise, (c) power at each frequency (Hz) for pink noise.

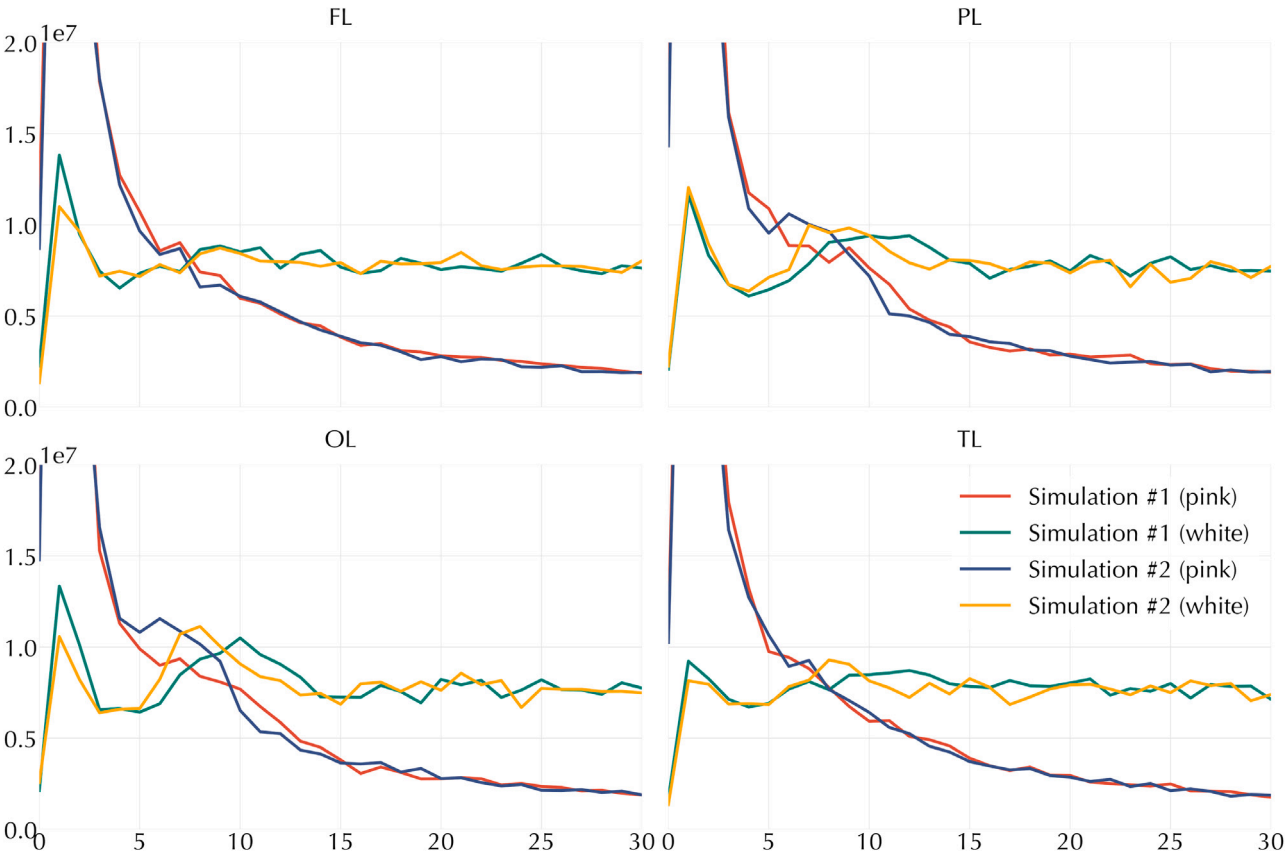


Fig. 12. Power plots of brain regions for the simulations.

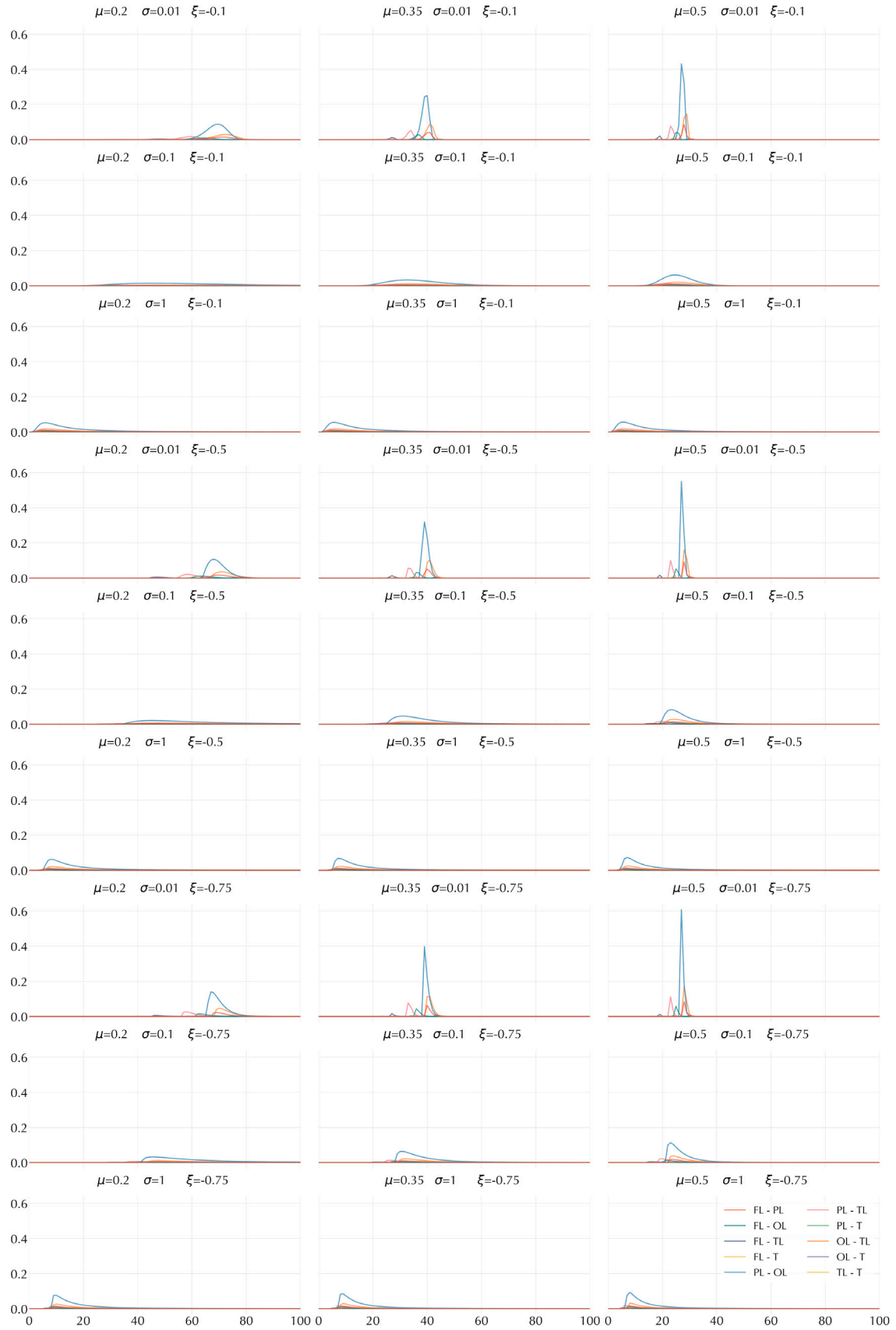


Fig. 13. Lag effects for different hypothetical distributions of axon diameters.



Fig. 14. Power spectra of VAR simulated signals for different hypothetical distributions of axon diameters.

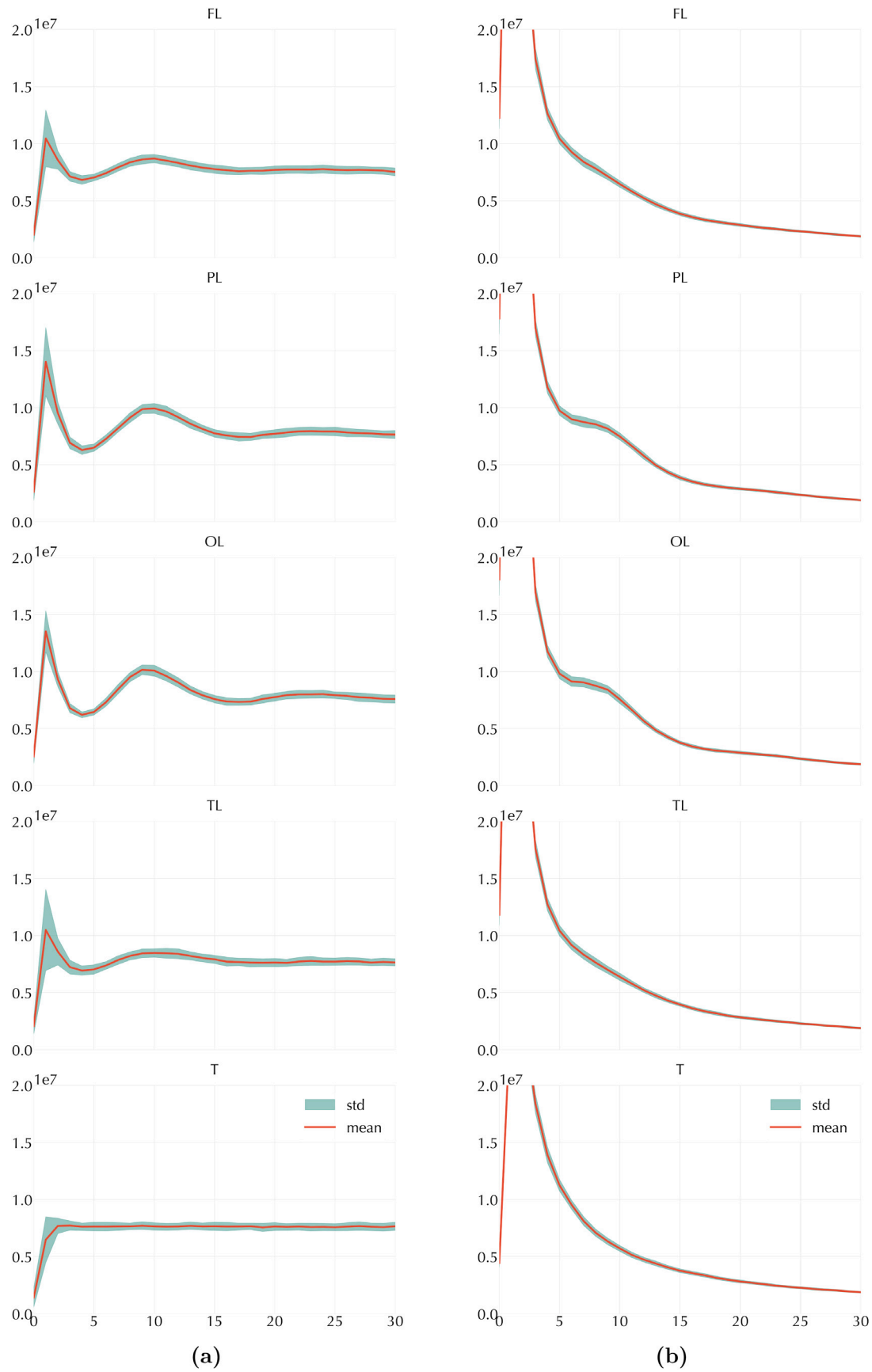


Fig. 15. Means and standard deviations in powers for the brain regions across 100 simulations with simulation #1 parameters with (a) white noise and (b) pink noise.

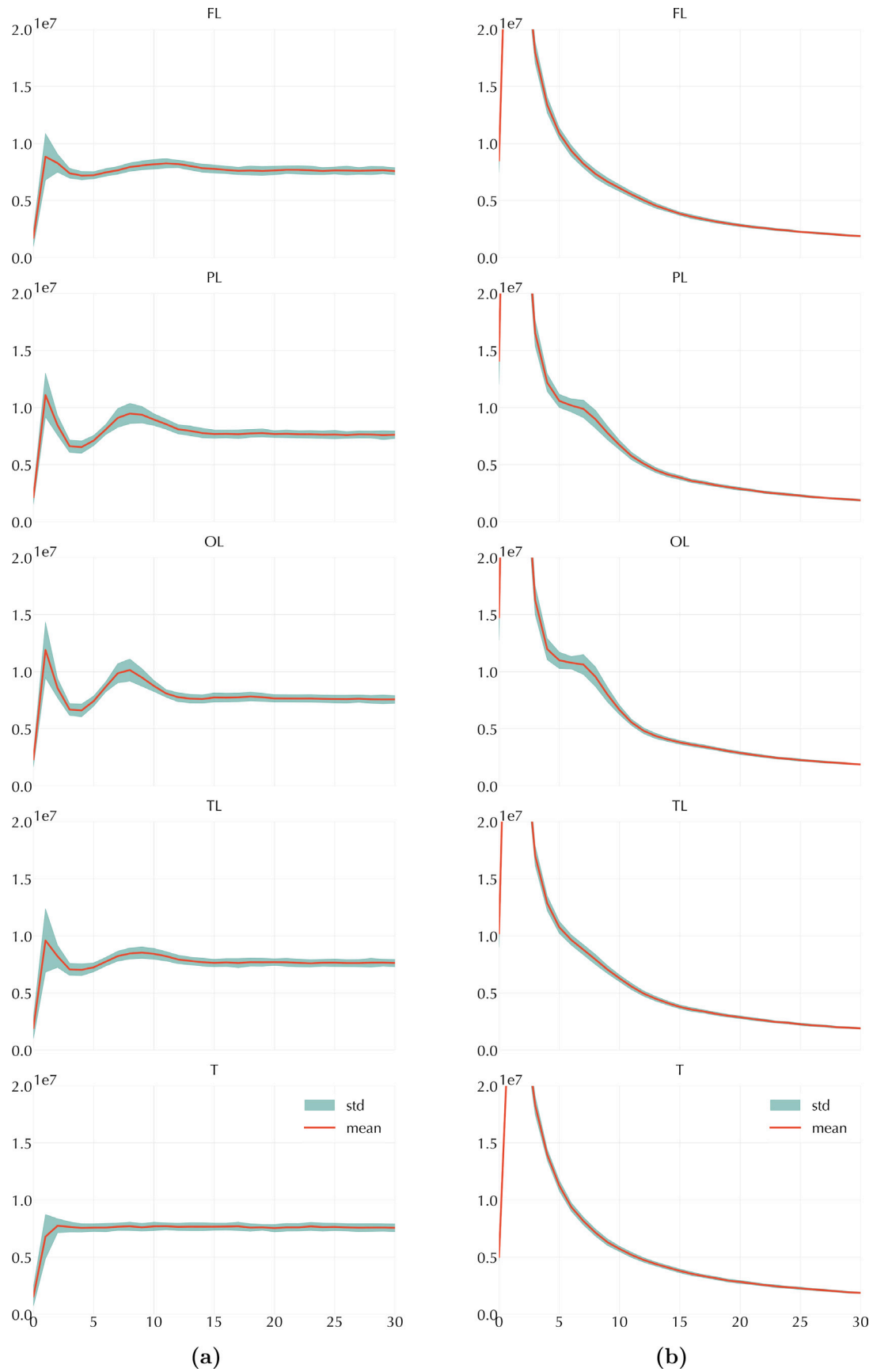


Fig. 16. Means and standard deviations in powers for the brain regions across 100 simulations with simulation #2 parameters with (a) white noise and (b) pink noise.

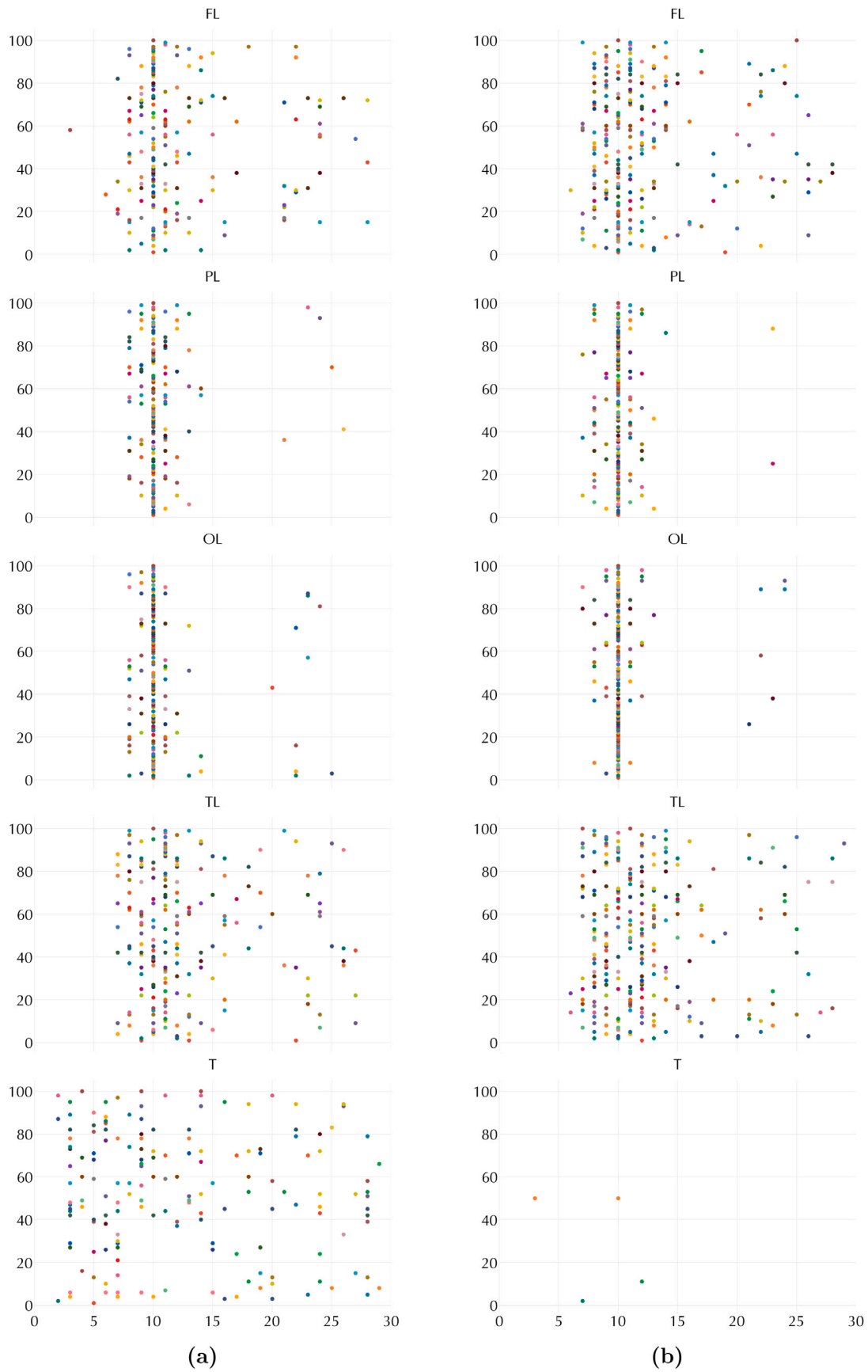


Fig. 17. Peak presence for the brain regions across 100 simulations with simulation #1 parameters with (a) white noise and (b) pink noise. Each dot represents the presence of a spectral peak at a specific frequency (x-axis) in a given simulation index (y-axis). Peaks were identified using FOOOF with a fixed aperiodic mode and 2–8 Hz peak width limits.



Fig. 18. Peak presence for the brain regions across 100 simulations with simulation #2 parameters with (a) white noise and (b) pink noise. Each dot represents the presence of a spectral peak at a specific frequency (x-axis) in a given simulation index (y-axis). Peaks were identified using FOOOF with a fixed aperiodic mode and 2–8 Hz peak width limits.



Fig. 19. The percentage of simulations (across 100 simulations per type) that contain at least one detected alpha peak for each brain region and simulation type. Simulation types are abbreviated as #1w (simulation #1 with white noise), #1p (simulation #1 with pink noise), etc.



Fig. 20. The percentage of all detected peaks that are located within the alpha band for each brain region and simulation type. Simulation types are abbreviated as #1w (simulation #1 with white noise), #1p (simulation #1 with pink noise), etc.

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